

Incident Handler's Journal

Date	31 May 2026
Entry	#1
Description	Documenting a ransomware security incident involving a health care organisation and analysing the cause, impact, and incident details using the 5 W's method
Tool(s) Used	None
The 5 W's	Who: An organised group of unethical hackers targeting organisations within the health care sector. What: A ransomware attack encrypted critical company files and disrupted access to medical records and business systems. When: The incident occurred on a Tuesday morning at approximately 9:00am. Where: The incident occurred within a small U.S. health care clinic specialising in primary-care services. Why: The incident occurred after employees received phishing emails containing malicious attachments. Once downloaded, malware installed ransomware that encrypted company files. The attackers appeared financially motivated and demanded payment in exchange for a decryption key.
Additional Notes	- How could stronger phishing awareness training and email filtering reduce the likelihood of future attacks? - Did the organisation keep secure backups and a disaster recovery plan to restore encrypted data without paying the ransom?

Date	3 June 2026
Entry	#2
Description	Analysing a packet capture file to understand network traffic and communication patterns. This activity improved my understanding of packet analysis and how network data helps investigations
Tool(s) Used	Wireshark was used to inspect packet capture data. It allows analysts to capture and analyse network traffic through a graphical interface and helps find suspicious communications
The 5 W's	N/A
Additional Notes	- I have previously used the interface before and found it easier to navigate with more practice

Date	5 June 2026
Entry	#3
Description	Capturing and filtering network traffic using a command line network protocol analyser. This activity strengthened my understanding of packet capture techniques and investigative workflows
Tool(s) Used	Tcpdump was used to capture and analyse network traffic. The tool supports packet capture, filtering, and troubleshooting from the command line
The 5 W's	N/A
Additional Notes	- Learning the command syntax was challenging but improved my confidence with Linux tools

Date	7 June 2026
Entry	#4

Description	Investigating a suspicious file hash reported by security monitoring systems. This activity occurred during the Detection and Analysis phase because additional investigation was needed to find whether the alert was a genuine threat
Tool(s) Used	VirusTotal was used to investigate a suspicious SHA-256 file hash. The platform compares indicators against multiple security vendors and threat intelligence sources
The 5 W's	Who: Unknown malicious actor. What: A malicious email attachment containing malware. When: An alert was generated after detection by security systems. Where: An employee workstation at a financial services company. Why: An employee downloaded and executed a malicious attachment received through email.
Additional Notes	- Could stronger email filtering and security awareness training reduce the likelihood of similar incidents?

Reflections / Notes

The tcpdump activity was the most challenging because I had limited experience with this specific tool. Learning the syntax took time and I made many mistakes before capturing the correct traffic. Repeating the activity improved my understanding and confidence.

My understanding of incident detection and response has improved significantly. I now better understand the incident response lifecycle, including detection, analysis, containment, recovery, and post-incident activities. The course demonstrated how people, processes, and tools work together during investigations.

I enjoyed learning about network traffic analysis with Wireshark and tcpdump. Being able to inspect communications between systems made cybersecurity concepts feel practical and engaging. These activities increased my interest in network security and incident investigation.